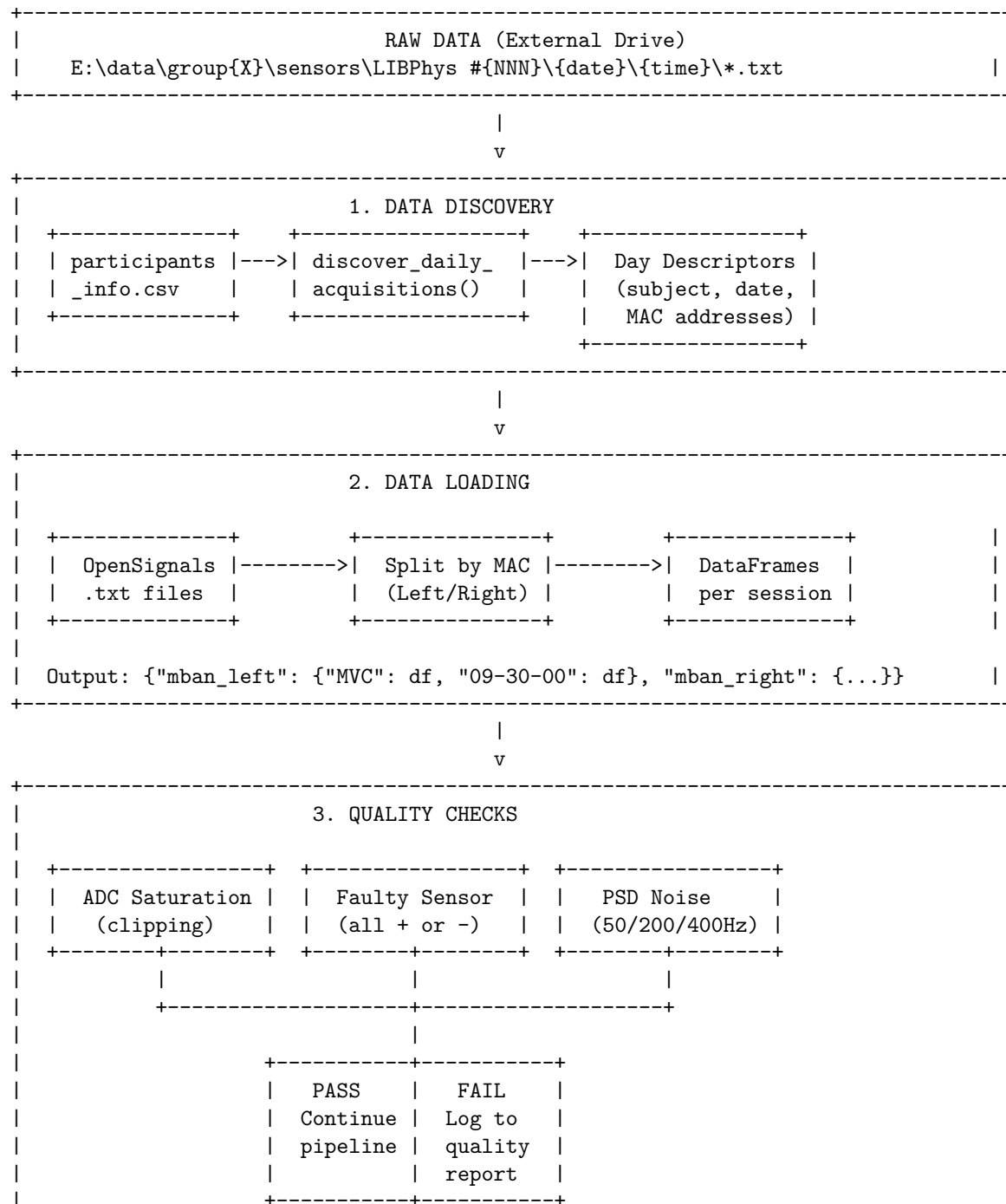


PrevOccupAI+ EMG Pipeline Overview

Purpose

Analyze surface EMG from trapezius muscles to assess occupational workload and generate personalized Occupational Health (OH) profiles.

High-Level Architecture



v				
CLASSIFY EACH 5-SECOND BIN				
[GREEN]	Below usual	< P10	"Light work for you"	
[GREEN]	Typical-low	P10 - P50	"Normal low intensity"	
[ORANGE]	Typical-high	P50 - P90	"Normal high intensity"	
[RED]	High for you	> P90	"Unusually intense"	
v				
7. OUTPUTS				
CSV Tables		OH Profiles (JSON)		Visualizations
* session_metrics			* APDF curves	
* daily_metrics		Per-subject file	* Histograms	
* weekly_metrics		with all EMG	* Timeline plots	
* quality_report		metrics stored	* Donut charts	

Code Organization

```

PrevOccupAI_Plus/
|
+-- main_emg.py                                <-- ENTRY POINT
|
+-- sensors/                                    <-- ALL EMG CODE
|   |
|   +-- emg_pipeline.py                        <-- Main orchestrator (run_emg_pipeline)
|   |
|   +-- load/                                  <-- Data Discovery & Loading
|       +-- dataset_loader.py                  * discover_daily_acquisitions()
|       +-- daily_data_loader.py               * load_daily_acquisitions()
|       +-- path_handler.py                    * File path resolution
|       +-- data_quality.py                     * Quality report structures
|
|   +-- process/                               <-- Signal Processing
|       +-- emg_preprocessing.py                * Filtering, envelope, MVC RMS
|       +-- emg_mvc.py                          * MVC segment detection
|       +-- emg_quality_analysis.py             * ADC saturation, PSD noise checks
|
|   +-- metrics/                               <-- Metric Computation
|       +-- emg_metrics.py                      * APDF, rest metrics, aggregations
|       +-- emg_output.py                       * CSV export, table building
|

```

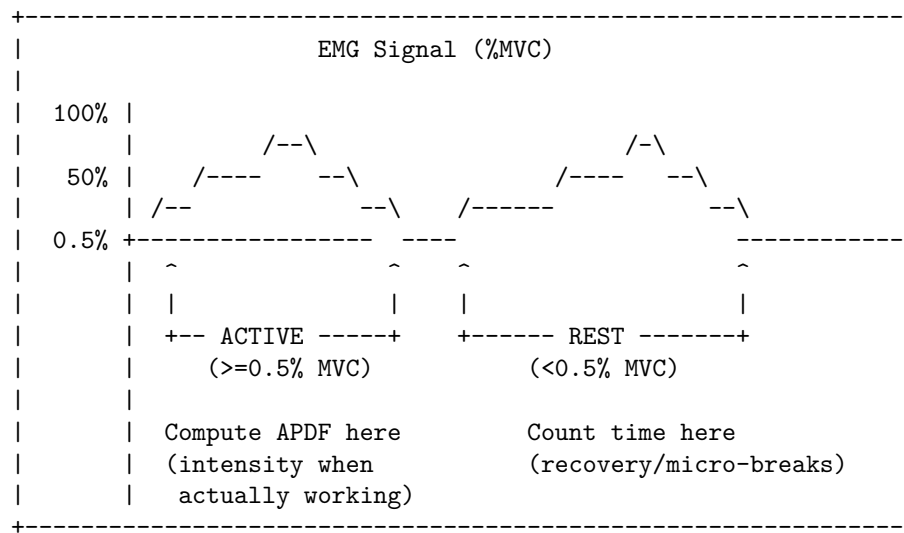
```

|   +-- visualize/                <-- Plotting
|       +-- emg_research.py        * Signal-based plots (APDF, timeline)
|       +-- emg_oh.py              * OH profile-based plots (donuts)
|
+-- OH_profile/                    <-- JSON Profile Management
    +-- emg_oh_helper.py           * Save EMG metrics to JSON
    +-- constants.py               * Metric key definitions

```

Key Scientific Framework

Active APDF + Rest Time (Veiersted et al., 2013)

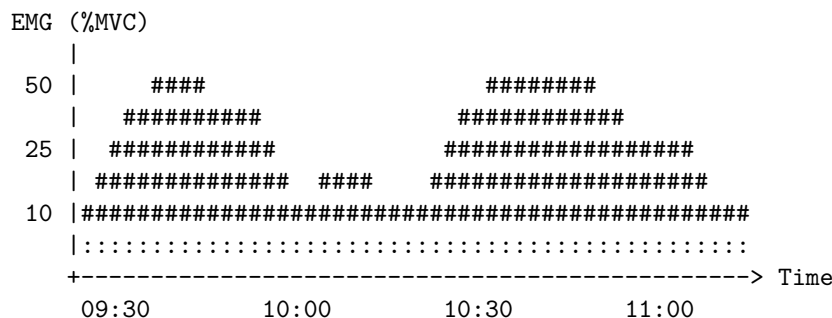


Key insight: Traditional APDF mixes “intensity when working” with “rest time”, making it hard to interpret. By separating them:

- **Active APDF** answers: “When this person is actually working, how hard?”
 - **Rest metrics** answer: “How much recovery time are they getting?”
-

Example Output: Session Timeline

Subject 81 - Left Trapezius - 2024-09-30 09:30



Legend:

High for you (>P90) :: Rest (<0.5% MVC)

```

## Typical-high (P50-P90)  -- P50 baseline
## Typical-low (P10-P50)   -- P10 baseline
## Below usual (<P10)

```

Clinical Relevance

Metric	What it tells us	Risk indicator
Active APDF P50	Typical muscle load when working	High = sustained strain
Rest %	Recovery time during work	Low = insufficient breaks
Gap frequency	Number of micro-pauses/min	Low = continuous tension
High-for-you %	Time above personal P90	High = overexertion risk

Running the Pipeline

```

# In main_emg.py
from main_emg import main

# Process all subjects
main(run_all=True)

# Process specific subjects
main(run_all=True, subject_filter=["81", "82", "83"])

# Quick test with limited data
main(run_all=True, max_subjects=2, max_days_per_subject=1)

```

References

- Veiersted, K. B., et al. (2013). “Assessment of occupational muscle activity.” *Applied Ergonomics*.
 - Hansson, G. A., et al. (2010). “Validity and reliability of triaxial accelerometers for inclinometry in posture analysis.” *Medical and Biological Engineering and Computing*.
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